Active Entity Ontology for Science (AEOS)

A portable, versioned model for scientific entities and interaction motifs

Active Inference Institute

2026-07-17

Publication note

This maintained repository source turns the AEOS conceptual framework into a portable, versioned, and locally validated ontology model. It is intended to support forks, downstream tools, and future publication revisions.

The complete 2022/v2 binary source snapshot is preserved in `../archive/2022-v2/`. The published work is identified by DOI 10.5281/zenodo.7484994.

Abstract

The Active Entity Ontology for Science (AEOS) describes scientific ecosystems as networks of active and informational entities. Active entities select policies and enact affordances, while informational entities contain or represent the resources acted upon. AEOS organizes these concepts through entity classes, entity types, roles, affordances, internal/sense/active states, and explicit interaction motifs.

This repository provides a source-controlled representation of those concepts in `data/aeos.yaml`, a JSON Schema contract, and a deterministic JSON export. The design keeps relationships explicit so that forks and downstream systems can validate and render AEOS without depending on an external interactive application.

Model

Entity classes and types

AEOS has two entity classes: **Active Entity** and **Informational Entity**. Active entities include humans, teams, organizations, computer systems, DAOs,

communities, laboratories, funding agencies, publishers, and knowledge markets. Informational entities include knowledge artifacts, papers, datasets, grants, code, intellectual property, assets, metadata, blockchains, cryptographic identities, smart contracts, and project catechisms.

Each type belongs to exactly one class. The complete versioned list and each definition are maintained in `../data/aeos.yaml`.

States, roles, and affordances

An active entity is modeled through internal states, incoming sense states, and outgoing active states. Roles describe contextual responsibilities. Affordances describe possible actions and declare their actors, targets, informational objects, and areas of concern.

The six current areas of concern are funding, communication, cryptographic relationships, scientific review, research and analysis, and publishing.

Interaction motifs

Motifs are ordered examples of ecosystem coordination. The source includes centralized and decentralized motifs for funding, communication, scientific review, research and analysis, and publishing. Each step names an actor, affordance, target, and optional informational object. This makes the examples machine-checkable while preserving the natural-language intent of the original publication.

Reproducible use

```
python3 -m pip install -e '.[test]'
aeos validate
aeos build
aeos build --check
```

The generated `../aeos.json` contains the source collections and a derived graph of typed nodes and edges. See `../docs/data-contract.md` for the stable contract.

Attribution

This work is released under CC BY 4.0. Derived works should credit the Active Inference Institute, identify their AEOS source release, and link to the publication DOI.